DenseRet: Learning Dual-Encoder Representations for Open-Domain Passage Retrieval

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Abstract

Open-domain question answering requires retrieving useful evidence from large collections of text before an answer can be produced. Traditional sparse retrieval matches questions and passages through shared words, which can fail when relevant evidence uses different terminology or paraphrases the question. We introduce DenseRet, a neural passage-retrieval approach based on a Dual Encoder. DenseRet uses one transformer encoder for questions and another for passages, mapping both into a shared vector space. The encoders are trained with Contrastive Learning so that questions and relevant passages are close together while unrelated passages are separated. We evaluate DenseRet on Natural Questions for Open-domain QA and Passage Retrieval. DenseRet substantially improves retrieval quality over a keyword-search baseline, particularly at a small retrieval depth. The results indicate that dense representations can capture semantic relationships that sparse matching misses, while maintaining efficient approximate nearest-neighbor search at inference time. These findings support dual-encoder retrieval as a practical foundation for scalable question answering systems.

Keywords

Dense retrieval, Dual Encoder, Contrastive Learning, Natural Questions, Open-domain QA, Passage Retrieval

1. Introduction

Open-domain QA systems must identify relevant evidence from a broad text collection before generating or selecting an answer. This makes Passage Retrieval a central component of reliable Open-domain QA. Keyword-based retrieval is efficient, but it depends heavily on lexical overlap and may overlook passages that express the same information with different wording.

DenseRet addresses this limitation by learning a shared semantic space for questions and passages. Its central design is a Dual Encoder: one encoder represents the question, while a second encoder represents candidate passages. Contrastive Learning then encourages relevant question-passage pairs to receive similar representations and unrelated pairs to receive dissimilar representations. Unlike sparse matching, this formulation can recognize semantic correspondence even when exact words differ.

We evaluate DenseRet on Natural Questions using retrieval recall at two operating points. The study focuses on whether learned representations improve the chance that an answer-bearing passage appears among the highest-ranked candidates. As a foundational paper, this work presents the method and evaluation without relying on prior cited work.

2. Method

DenseRet receives a natural-language question and a collection of candidate passages. A question encoder converts the question into a fixed-dimensional vector, while a passage encoder independently converts each passage into a vector of the same dimension. Similarity is computed between the question vector and each passage vector using normalized dot product. At serving time, passage vectors can be computed in advance, allowing a new question to be matched efficiently against the indexed collection.

Training uses paired questions and relevant passages together with passages that are not associated with the question. Under Contrastive Learning, the score of the relevant pair is increased while the scores of competing passages are

decreased within the same training batch. This objective teaches the Dual Encoder to preserve information useful for retrieval rather than memorizing isolated lexical matches.

The two encoders use transformer-based sequence representations with separate parameters. The question and passage inputs are truncated to fixed limits, and their final representations are projected into a shared retrieval space. DenseRet ranks passages by vector similarity. The resulting ranked list is passed to downstream Open-domain QA components, which can inspect the retrieved evidence and produce an answer. This separation keeps retrieval independently measurable through Passage Retrieval metrics.

3. Experiments

We evaluate DenseRet on Natural Questions, measuring whether an answer-bearing passage is retrieved within the top 20 or top 100 results. We compare against a keyword-search baseline using the same passage collection and evaluation questions. The primary metric is Recall@k, which records the percentage of questions for which at least one relevant passage appears among the first k retrieved items.

DenseRet improves Recall@20 by 9.1 points over the baseline, showing its greatest advantage when the retrieval budget is small. The improvement remains positive at Recall@100, although the gap narrows as both systems are allowed to return more passages.

System	Recall@20	Recall@100
Keyword-search baseline	39.6	64.2
DenseRet	48.7	71.8

4. Conclusion

DenseRet presents a simple dense-retrieval formulation for Open-domain QA. By combining a Dual Encoder with Contrastive Learning, it places semantically related questions and passages near one another in vector space. On Natural Questions, DenseRet improves both Recall@20 and Recall@100 over a keyword-search baseline, with a particularly strong gain at shallow retrieval depth. These results suggest that learned passage representations are a useful foundation for scalable question answering.

References

References: None.